

DATA COMMUNICATION & NETWORKING

Comprehensive Study Notes

Chapter 03 & Chapter 05

Sine Wave

Wavelength λ

Phase θ

ASK

FSK

PSK

QAM

ETE 2026

Engineering Telecommunications

Chapters 03 & 05
Signals & Modulation

Table of Contents

Chapter 03 — Signals

- 1 Sine Wave
- 2 Wavelength (λ)
- 3 Phase (θ)
- 4 Composite Signals

Chapter 05 — Modulation Techniques

- 5 Modulation & Carrier Signal
- 6 Bit Rate and Baud Rate
- 6a Baud Rate & Bandwidth Relationship
- 7 Amplitude Shift Keying (ASK)
- 8 Frequency Shift Keying (FSK)
- 9 Phase Shift Keying (PSK)
- 10 Quadrature Amplitude Modulation (QAM)

Σ Summary & Quick Reference

- Modulation Techniques — Master Comparison
- All Key Formulas
- Final Summary

CHAPTER 03

Signals

Sine Waves · Wavelength · Phase · Composite Signals

1 Sine Wave

A **sine wave** is the most fundamental form of a periodic (repeating) signal. It represents a smooth oscillation and serves as the basic building block for all complex signals. Any real-world signal — voice, music, data — can be mathematically decomposed into a sum of sine waves (**Fourier Analysis**).

MATHEMATICAL DEFINITION

$$S(t) = A \cdot \sin(2\pi ft + \theta)$$

Symbol	Name	Meaning	Unit
$S(t)$	Instantaneous Amplitude	Signal value at time t	Volts (V)
A	Peak Amplitude	Maximum height of wave	Volts (V)
f	Frequency	Cycles per second	Hertz (Hz)
t	Time	Time variable	Seconds (s)
θ	Phase	Starting horizontal position	Degrees° / Radians

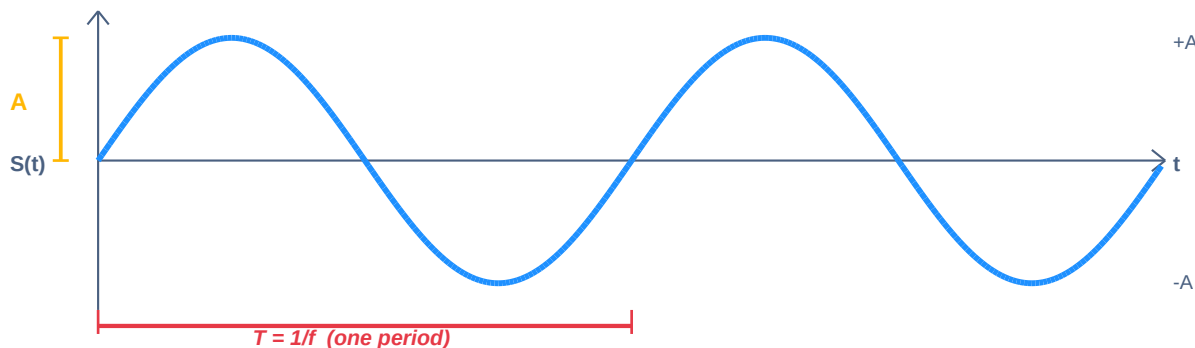


Fig 1.1 — Sine wave $S(t) = A \cdot \sin(2\pi ft + \theta)$. Peak height = Amplitude A . Red bracket = one complete period T .

The 3 Defining Parameters of a Sine Wave

- **Amplitude (A)** → Controls the POWER / STRENGTH of the signal (peak voltage).
- **Frequency (f)** → Controls how FAST the wave oscillates (cycles per second).
- **Phase (θ)** → Controls WHERE the wave starts on the time axis (horizontal shift).

Period and Frequency

Period **T** is the time to complete one full cycle. Frequency **f** is the number of cycles per second. They are exact reciprocals: **T = 1/f** and **f = 1/T**.

Time Unit	Equivalent	Frequency Unit	Equivalent
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz

Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

2 Wavelength (λ)

Wavelength (λ) is the physical distance a signal travels during one complete cycle. For electromagnetic waves in free space:

WAVELENGTH FORMULA

$$\lambda = C / f \quad \text{where } C = 3 \times 10^8 \text{ m/s}$$

🔪 SOLVED EXAMPLE — Wavelength Calculation

Problem: Find the wavelength in vacuum for a frequency of $4 \times 10^{14} \text{ Hz}$.

Given: $f = 4 \times 10^{14} \text{ Hz}$, $C = 3 \times 10^8 \text{ m/s}$

Solution: $\lambda = C / f = (3 \times 10^8) \div (4 \times 10^{14}) = 0.75 \times 10^{-6} \text{ m} = 0.75 \mu\text{m}$

→ This wavelength falls in the near-infrared light spectrum.

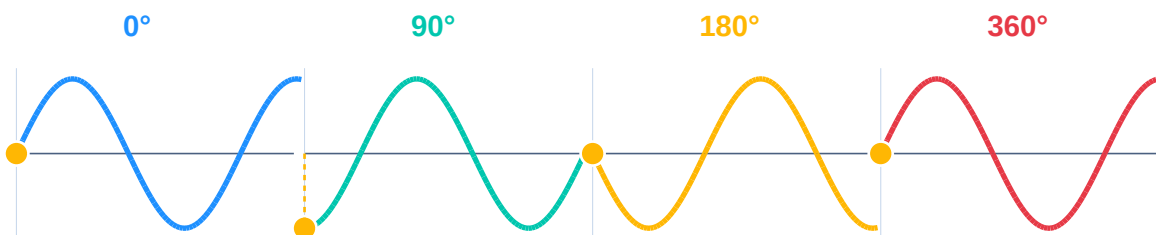
★ Key Facts — Wavelength

- Higher frequency → shorter wavelength (inverse relationship: $\lambda \propto 1/f$)
- Radio waves: $\lambda \approx 1 \text{ cm} - 10 \text{ km}$ | Visible light: $\lambda \approx 400 - 700 \text{ nm}$
- Wavelength determines antenna size — practical antenna length $\approx \lambda/4$

3 Phase (θ)

Phase describes the position of a sine wave relative to a reference point in time. It measures the horizontal shift of the wave. Phase is expressed in degrees ($^\circ$) or radians (rad). One full cycle = $360^\circ = 2\pi \text{ rad}$.

Phase Shifts of a Sine Wave — 0° , 90° , 180° , 360°



Gold dot = starting position of wave at $t=0$

Fig 3.1 — Each panel shows a sine wave shifted by the stated phase. The gold dot marks the starting position ($t = 0$). 360° is identical to 0° — one full cycle brings the wave back to its start.

🔪 SOLVED EXAMPLE — Phase Calculation

Problem: A sine wave is offset by $1/6$ of a cycle. Find the phase in degrees and radians.

Step 1: 1 full cycle = $360^\circ \rightarrow 1/6 \text{ cycle} = 360 \div 6 = 60^\circ$

Step 2: Convert: $60^\circ \times (2\pi / 360) = 1.047 \text{ radians} \approx \pi/3 \text{ rad}$

4 Composite Signal

Composite signals are combinations of many simple sine waves, each with different amplitudes, frequencies, and phases. Real data communication cannot use a single sine wave because it carries no information — information requires variation over time.

Application	Signal Type	Details
Power Distribution	Single sine wave	60 Hz — power companies to homes/offices
Burglar Alarm	Single sine wave	Simple alert signal triggered by sensors
Data Communication	Composite signal	Many sine waves combined — required to carry data

■ Composite Signals — Fourier Principle

- Fourier Theorem: ANY complex signal can be decomposed into pure sine waves.
- Fourier Synthesis: Combining sine waves builds complex signals.
- Bandwidth of a composite signal = highest frequency – lowest frequency.

CHAPTER 05

Modulation Techniques

ASK · FSK · PSK · QAM

5 Modulation

Modulation is the process of changing one or more properties (amplitude, frequency, or phase) of a high-frequency **carrier signal** based on a lower-frequency message/data signal. The resulting modulated signal can be efficiently transmitted through a medium.

■ Why Modulation is Necessary

1. Digital signals cannot travel long distances without severe degradation.
2. Analog (sinusoidal) carrier waves propagate efficiently through air, cables, and fiber.
3. Modulation enables multiplexing — multiple signals sharing the same channel.
4. Matches signal bandwidth to the physical medium (e.g., specific radio frequency bands).

Carrier Signal

A **carrier signal** is a high-frequency, unmodulated sine wave produced by the transmitter. It acts as the "vehicle" carrying information. The carrier itself contains no data — its parameters are altered by modulation to encode data. Carrier frequency is always much higher than the message signal frequency (typically 10× or more).

6 Bit Rate and Baud Rate

Term	Definition	Unit
Bit Rate	Total number of bits transmitted per second	bps (bits/second)
Baud Rate	Number of signal elements (symbols) sent per second	Baud
Relationship	$\text{Baud Rate} = \text{Bit Rate} \div \text{Bits per Signal Element}$	$\text{Bit Rate} \geq \text{Baud Rate}$

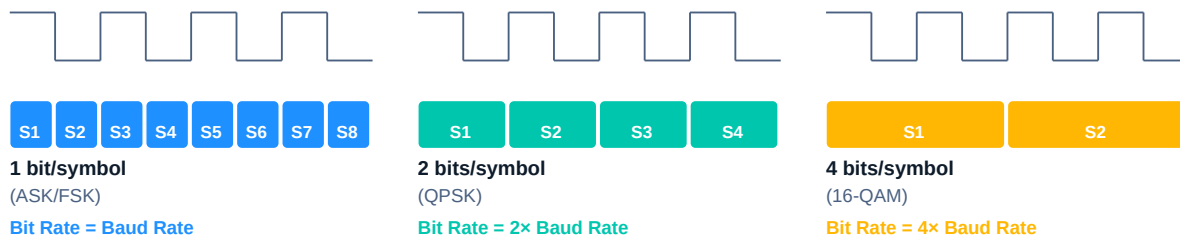


Fig 6.1 — As bits-per-symbol increases, more data fits into the same number of transmitted symbols — the baud rate stays constant while bit rate multiplies.

■ Bit Rate vs Baud Rate — Important Rules

- 1 bit/symbol → Bit Rate = Baud Rate (e.g., basic ASK / FSK)
- 2 bits/symbol → Bit Rate = 2 × Baud Rate (e.g., QPSK)
- 4 bits/symbol → Bit Rate = 4 × Baud Rate (e.g., 16-QAM)
- Baud rate determines how much BANDWIDTH is needed.
- More bits/symbol = same bandwidth, higher data speed.

6a Baud Rate & Bandwidth Relationship

The **bandwidth (BW)** required by an ASK signal depends directly on the **baud rate** and a roll-off factor **d** ($0 \leq d \leq 1$) that accounts for practical filter shaping:

ASK BANDWIDTH FORMULA

$$BW = (1 + d) \times N_{\text{baud}}$$

MINIMUM BANDWIDTH (D = 0)

$$BW_{\text{min}} = N_{\text{baud}}$$

d value	Interpretation	Formula	Result for N = 2000 Baud
d = 0	Ideal / minimum BW	$BW = 1 \times N_{\text{baud}}$	2,000 Hz
d = 0.5	Typical / moderate filter	$BW = 1.5 \times N_{\text{baud}}$	3,000 Hz
d = 1	Maximum / real-world BW	$BW = 2 \times N_{\text{baud}}$	4,000 Hz

Baud Rate vs Bandwidth — Graphical Relationship

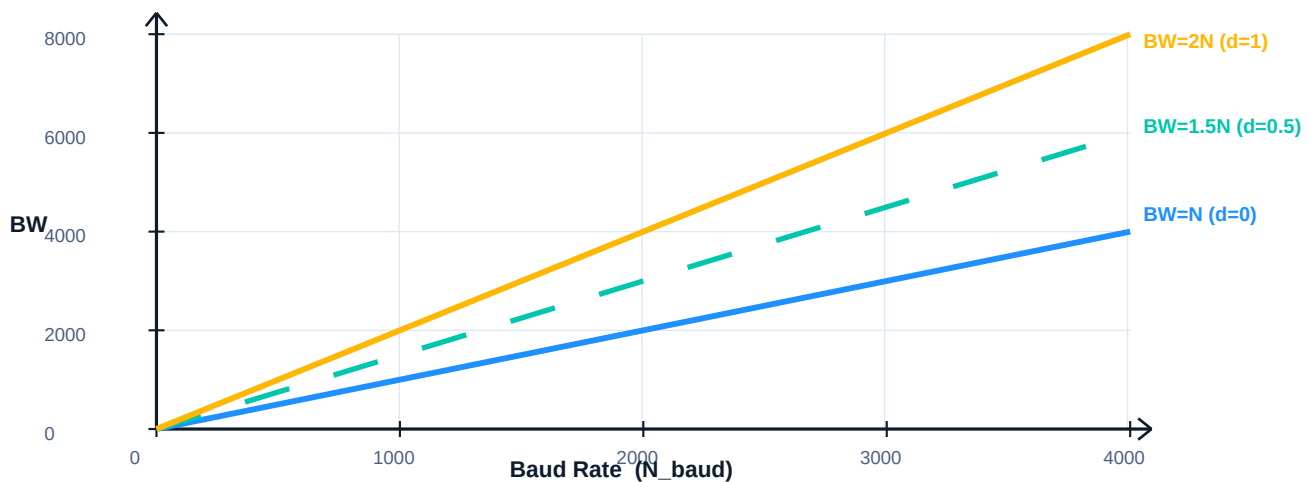


Fig 6a.1 — BW vs Baud Rate. Blue solid ($d=0$): $BW = N_{\text{baud}}$. Teal dashed ($d=0.5$): $BW = 1.5 \times N_{\text{baud}}$. Gold ($d=1$): $BW = 2 \times N_{\text{baud}}$. All three are strictly linear.

✍ SOLVED EXAMPLE — Full-Duplex ASK Design

Problem: Total BW = 10,000 Hz (from 1,000 to 11,000 Hz). Design a full-duplex ASK system.

Step 1: Full-duplex splits the band into 2 equal halves:

BW per direction = $10,000 \div 2 = 5,000$ Hz

Step 2: Carrier = centre of each sub-band:

Forward: $f_c = 1,000 + 5,000/2 = 3,500$ Hz [band: 1,000 – 6,000 Hz]

Backward: $f_c = 6,000 + 5,000/2 = 8,500$ Hz [band: 6,000 – 11,000 Hz]

★ Key Takeaways — Baud Rate & BW

- BW is directly proportional to baud rate — doubling baud rate doubles required BW.
- The factor d is a design parameter; real systems use $0 < d \leq 1$.
- To reduce BW consumption: use higher-order modulation (more bits/symbol).
- Same baud rate + higher bits/symbol = same BW but higher bit rate.

7 Amplitude Shift Keying (ASK)

ASK encodes binary data by changing the **amplitude** of the carrier while keeping frequency and phase constant. High amplitude = binary 1; low/zero amplitude = binary 0.

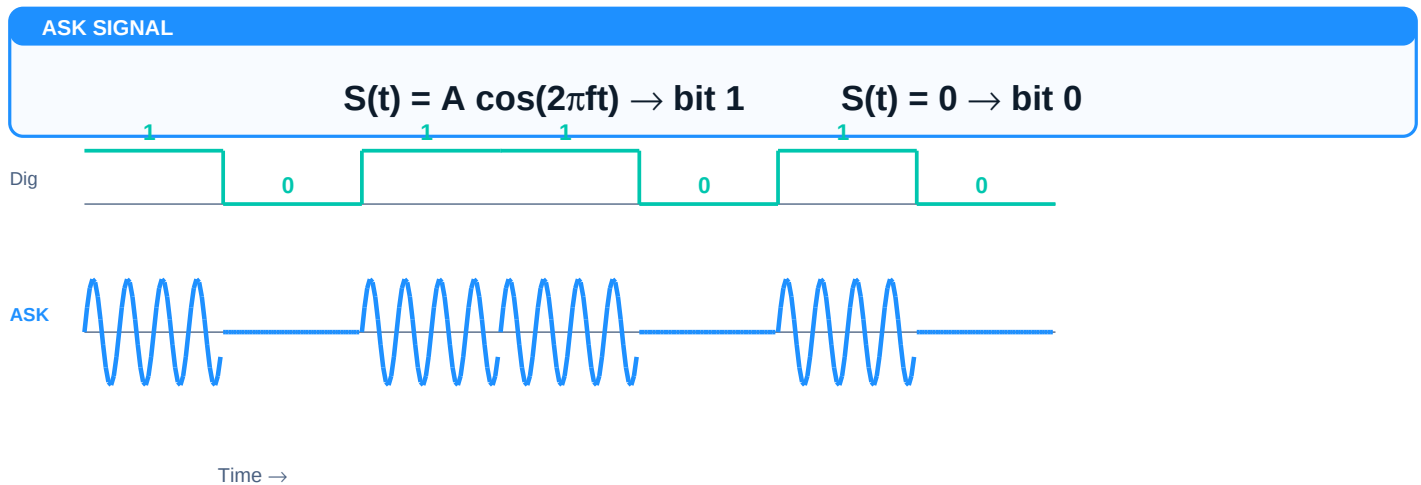


Fig 7.1 — ASK waveform: high amplitude = bit 1, zero amplitude = bit 0. Frequency and phase remain constant.

Property	Detail
Noise Susceptibility	HIGH — amplitude fluctuations directly mimic data changes
Special Variant	OOK (On-Off Keying): zero voltage = 0, reduces energy use
Efficiency	Low — poor bandwidth utilization
Application	Transmit digital data over optical fiber
Max Speed	Up to 1,200 bps on voice-grade lines
Bandwidth	$BW = (1 + d) \times N_{\text{baud}}$ ($BW_{\text{min}} = N_{\text{baud}}$ when $d = 0$)

8 Frequency Shift Keying (FSK)

FSK encodes binary data by changing the **frequency** of the carrier while keeping amplitude and phase constant. High frequency = binary 1; low frequency = binary 0.

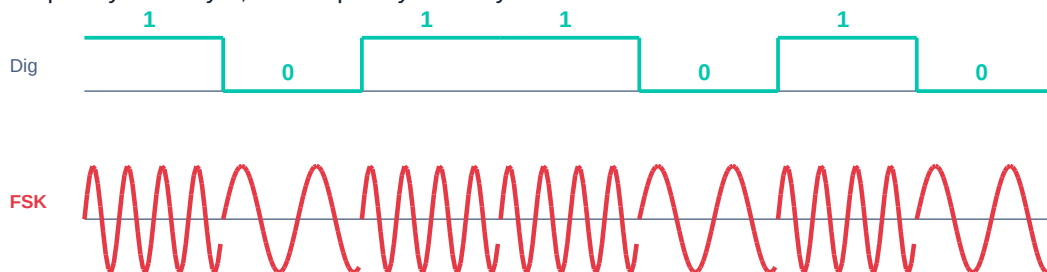


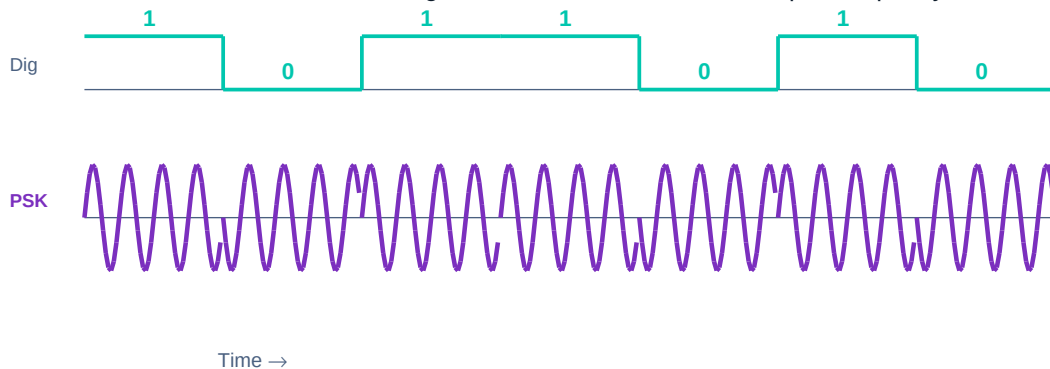
Fig 8.1 — FSK waveform: high frequency = bit 1, low frequency = bit 0. Amplitude and phase remain constant.

Property	Detail
Noise Resistance	Better than ASK — amplitude noise does not affect frequency

Voltage Spikes	Ignored — not amplitude-based detection
Max Speed	1,200 bps on voice-grade telephone lines
Full Duplex	Uses two separate frequency sub-bands simultaneously
Application	Radio transmission, early dial-up modems, paging systems
Voice-Grade Range	300 Hz to 3,400 Hz (standard telephone lines)

9 Phase Shift Keying (PSK)

PSK encodes binary data by changing the **phase** of the carrier while keeping amplitude and frequency constant. BPSK uses 0° for bit 0 and 180° for bit 1. Higher-order variants encode multiple bits per symbol.



Time →
Fig 9.1 — PSK waveform: 180° phase reversal encodes bit transitions. Amplitude and frequency remain constant.

Variant	Phases Used	Bits/Symbol	Throughput
BPSK (Binary)	2 (0° , 180°)	1 bit	Low — very robust
QPSK (Quadrature)	4 (0° , 90° , 180° , 270°)	2 bits	Doubles BPSK
8-PSK	8 phases	3 bits	Higher speed
16-PSK	16 phases	4 bits	Even higher speed

★ Key Facts — PSK

- PSK is more bandwidth-efficient than both ASK and FSK.
- Used in WiFi (802.11), Bluetooth, 4G/5G cellular networks, and satellite links.
- Higher phase counts → more bits per symbol but increased noise sensitivity.
- QPSK is a very popular balance between efficiency and robustness.

10 Quadrature Amplitude Modulation (QAM)

QAM combines ASK and PSK — it simultaneously varies both **amplitude and phase** of the carrier. This allows many more distinct signal states (symbols), encoding multiple bits per symbol and achieving very high data rates within limited bandwidth.

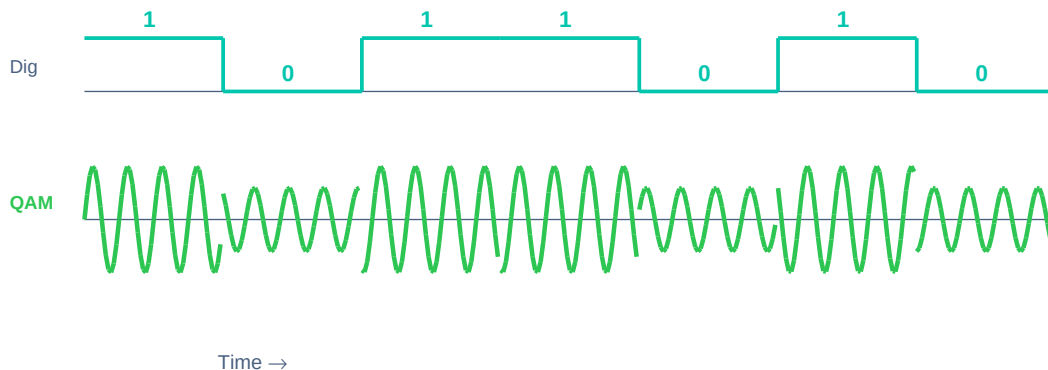


Fig 10.1 — QAM waveform: both amplitude and phase vary to encode bit patterns.

QAM Constellation Diagrams

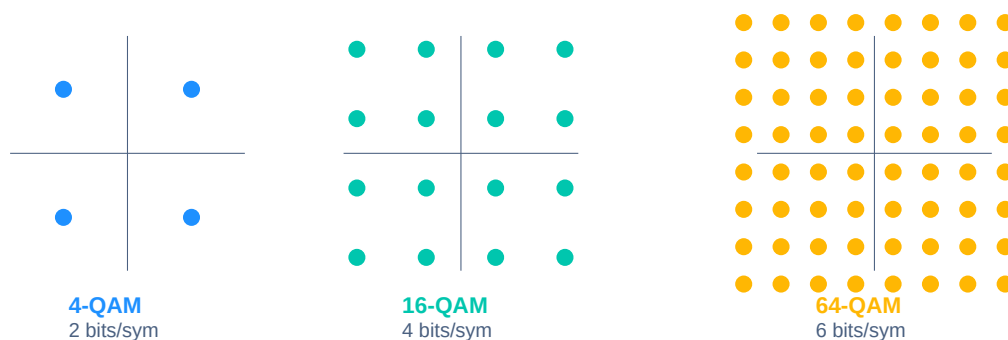


Fig 10.2 — Constellation diagrams. Each dot = one symbol. More dots = more bits/symbol = higher throughput in the same bandwidth.

QAM Type	Signal States	Bits/Symbol	Real-World Use
4-QAM / QPSK	4	2 bits	Satellite TV, basic WiFi
16-QAM	16	4 bits	4G LTE, DVB-T
64-QAM	64	6 bits	Cable TV, ADSL
256-QAM	256	8 bits	5G NR, DOCSIS 3.1
1024-QAM	1024	10 bits	Advanced 5G, fiber-like speeds

■ QAM — Power & Trade-offs

- QAM is the most bandwidth-efficient modulation technique.
- Higher QAM order = more bits per baud = faster data at the same bandwidth.
- Trade-off: Higher QAM is more sensitive to noise — requires cleaner channels.
- 256-QAM and 1024-QAM are used in modern 5G and cable broadband systems.

CHAPTER Σ

Summary & Quick Reference

All Formulas · Master Comparison · Key Rules

Modulation Techniques — Master Comparison

Feature	ASK	FSK	PSK	QAM
Parameter Changed	Amplitude	Frequency	Phase	Amplitude + Phase
Stays Constant	f and θ	A and θ	A and f	Frequency only
Noise Resistance	■ Poor	✓ Good	✓✓ Very Good	~ Moderate–Good
BW Efficiency	Low	Low	Medium	★ High
Complexity	Simple	Simple	Moderate	Complex
Max Speed (voice)	1,200 bps	1,200 bps	High	Very High
Used In	Optical fiber	Radio, modems	WiFi, 4G/5G	Cable, 5G, LTE

All Key Formulas

Formula	Expression
Sine Wave Equation	$S(t) = A \cdot \sin(2\pi ft + \theta)$
Period & Frequency	$T = 1/f \leftrightarrow f = 1/T$
Wavelength	$\lambda = C / f$ ($C = 3 \times 10^8$ m/s)
Baud Rate	Baud Rate = Bit Rate ÷ Bits per Symbol
ASK Bandwidth	$BW = (1 + d) \times N_{\text{baud}}$
Minimum Bandwidth	$BW_{\text{min}} = N_{\text{baud}}$ ($d = 0$)
BW vs Baud ($d=0.5$)	$BW = 1.5 \times N_{\text{baud}}$ (typical real-world)
BW vs Baud ($d=1$)	$BW = 2 \times N_{\text{baud}}$ (maximum)
Phase Conversion	$1^\circ = 2\pi/360$ rad $60^\circ = \pi/3 \approx 1.047$ rad
Cycle → Degrees	1 cycle = $360^\circ \rightarrow \text{fraction} \times 360 = \text{degrees}$

Final Summary — Everything to Remember

1. Sine wave has 3 parameters: **Amplitude (A)**, **Frequency (f)**, **Phase (θ)**.
2. Wavelength $\lambda = C/f$ — inversely proportional to frequency.
3. Modulation changes A, f, or θ of a carrier signal to encode data.
4. **ASK** = amplitude changes | **FSK** = frequency changes | **PSK** = phase changes.

5. **QAM** = both amplitude AND phase change simultaneously — highest bandwidth efficiency.
6. Bit Rate $\geq$ Baud Rate always. Higher-order modulation $\rightarrow$ more bits per baud.
7. Baud rate determines bandwidth; more bits/symbol = same BW, higher speed.
8. **BW = (1 + d) $\times$ N_baud** — ranges from N_baud (d=0) to 2 $\times$ N_baud (d=1).
9. QAM constellations: 4-QAM (2 bits) $\rightarrow$ 16-QAM (4) $\rightarrow$ 64-QAM (6) $\rightarrow$ 256-QAM (8) $\rightarrow$ 1024-QAM (10).